

GREENWHEEL DBT ASSIGNMENT

Analytics Engineering Training Workshop

1. The Problem Statement

The Scenario:

Welcome to **GreenWheel**, an urban electric bike-share startup. As our first Analytics Engineer, you have inherited a "Data Swamp." Raw CSV files are being uploaded to the warehouse, but leadership cannot use them for decision-making.

The Mission: You must build a robust dbt project that converts these raw files into a trusted "Marts" layer. You are specifically tasked with solving three issues:

- **Scale:** The trips table is huge. You must use **Incremental** loading to only process new data.
- **Data Quality:** Sensor glitches cause negative durations. You must use **dbt Tests** to flag these.
- **Auditability:** Prices change monthly. You must use **Snapshots** to track rental price history (SCD Type 2).

2. Data Preparation

Download the provided `raw_users.csv`, `raw_bikes.csv`, and `raw_trips.csv`. Place them in your `seeds/` directory and initialize the project.

```
dbt seed
```

3. The Transformation Layer

Model	Strategy	Instruction
stg_users	View	Use <code>{{ source() }}</code> . Clean names/emails and remove null records.
dim_bikes	Table	Use <code>{{ source() }}</code> . Create a logic to categorize bike tiers based on price.
fct_trips	Incremental	Use <code>{{ ref() }}</code> . Configure a unique key and a look-back window for new records.

4. YAML Configuration & Quality

Your project must include three YAML files to manage metadata and data quality:

- **seeds.yml**: Document columns and add `not_null` tests for IDs.
- **sources.yml**: Define the `greenwheel` source with a 24-hour freshness check.
- **marts.yml**: Apply unique, `not_null`, and relationships (foreign key) tests.

5. Advanced Components

Snapshots: Create `snapshots/bike_prices_snapshot.sql` using the `check` strategy on `rental_price`.

Macros: Create `macros/cents_to_dollars.sql` and apply it to the `trip_cost` in `fct_trips`.

Singular Test: Write `tests/assert_positive_duration.sql` to catch rows where `duration < 0`.

6. Submission Checklist

Run the following suite to validate your build:

```
dbt run && dbt snapshot
dbt test
dbt docs generate && dbt docs serve
```